

MID-TERM EXAMINATION

Course: **COMPUTER SYSTEM PROGRAMMING**

Time: **60 minutes**

Term: 1 – Academic year: **2021-2022**

Lecturer(s): **Dinh Dien**

Student name:

Student ID:

Problem 1. (20 points):

Assume we are running code on a 5-bit binary and 2-digit hexadecimal machine and we can represent a decimal number in 1's complement, 2's complement and bias 10. Fill in the empty boxes. Write number in 2-digit hexadecimal format. If the number can not be represented, please fill “_”.

Description	1's complement	2's complement	Bias 10
0			
-7			
16			
-14			
20			

Short explanation: _____

Problem 2. (20 points):

Assume a 12-bit floating-point representation (1 sign bit, 5 exponent bits, and 6 fraction bits). Use “round to nearest, half to even” (as in IEEE 754).

a. Convert 0 10010 111010 (a 12-bit FP) to its decimal equivalent.

The exponent is: _____

The value of the fraction (without implicit 1.) is: _____

Final decimal value: _____

b. Convert -8.2 to its 12-bit floating-point equivalent (rounding if needed).

The base-2 normal notation of -8.2 is: _____

The 5-bit encoding of the exponent is: _____

The 6-bit encoding of the fraction is: _____

c. Using the same 12-bit floating-point format (3 hex digits):

The number $-\infty$ (positive infinity) has hex encoding: _____

The largest denormalized number has hex encoding: _____

Short explanation: _____

Problem 3. (20 points):

In the following questions assume the variables a and b are signed integers and that the machine uses two's complement representation. Also assume that MAX_INT is the maximum integer, MIN_INT is the minimum integer, and W is one less than the word length (e.g., W=31 for 32-bit integers).

Match each of the descriptions on the left with a line of code on the right (write in the letter).

1. a&b	a. $\sim(\sim a \mid (b \wedge (\text{MIN_INT} + \text{MAX_INT})))$
2. a*7	b. $((a \wedge b) \& \sim b) \mid (\sim(a \wedge b) \& b)$
3. a	c. $1 + (a << 3) + \sim a$
4. !a	d. $(a << 4) + (a << 2) + (a << 1)$
5. (a<0)?1:-1	e. $a \wedge (\text{MIN_INT} + \text{MAX_INT})$
	f. $\sim((a \mid (\sim a + 1)) >> W) \& 1$
	g. $\sim((a >> W) << 1)$
	h. $a >> 2$

Short

explanation: _____

Problem 4.

Address	Value	Register	Value
0x100	0xFF	%eax	0x100
0x104	0xAB	%ecx	0x1
0x108	0x09	%edx	0x3
0x10C	0x11		

Fill in the following table the effect of the following instructions, both in term of the register or memory location that will be updated and the resulting value:

Instruction	Destination	Value
addl 8(%eax),%edx		
subl %edx,4(%eax)		
imull \$16,(%eax,%edx)		
addl %edx,%eax		

incl (%eax)		
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Short explanation: _____

Problem 5.

Complete the C code

<pre> /* ptr=%rdi, x=%esi */ int f2(int *ptr, int x) { int res = _____ ; for (int i = 0; _____ ; i++) { if (_____ < _____) { _____ = _____ ; } } return res; } </pre>	<pre> f2: movl %esi,%eax movl \$0,%ecx jmp .L2 .L3: addl \$1,%ecx .L2: movslq %ecx,%rdx movl (%rdi,%rdx,4),%edx testl %edx,%edx jg .L6 cmpl %eax,%edx jge .L3 movl %edx,%eax jmp .L3 .L6: ret </pre>
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Short explanation: _____